Dice Throws

AlgoExpert · Hard · Dynamic Programming

The problem

- Number of ways to roll numDice dice with numSides sides to total `target`.

Intuition

- Count ways to reach each total with dice added one at a time.

Approach

- 1. ways for total zero with no dice is one
- 2. For each die, roll every face
- 3. Shift counts forward
- 4. Read ways of target

Java solution

```java
public class DiceThrows {
    public static int diceThrows(int numDice, int numSides, int target) {
        long[] ways = new long[target + 1];
        ways[0] = 1;
        for (int d = 0; d < numDice; d++) {
            long[] next = new long[target + 1];
            for (int t = 0; t <= target; t++) if (ways[t] != 0)
                for (int s = 1; s <= numSides && t + s <= target; s++) next[t + s] += ways[t];
            ways = next;
        }
        return (int) ways[target];
    }
}
```

Complexity

- Time: O(d * t * s)
- Space: O(t)

Recap

- Count ways to reach each total with dice added one at a time.
- Complexity: O(d * t * s)